The space Y_2 is not hereditarily Dunford–Pettis

Full negative answer to Question 3.5(1) of arXiv:1302.6369

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Status. `candidate_full_likely_valid`.

Abstract

Kalenda and Spurny asked whether their explicit c_0-sum Y_2 is hereditarily Dunford–Pettis. We answer no. A tail-diagonal map embeds the Pełczyński–Szlenk space Y_1, which fails the Dunford–Pettis property, as a closed subspace of Y_2. The embedding has distortion at most two and follows from a direct comparison between the full seminorms A_p defining Y_1 and the truncated norms defining the summands of Y_2.

1 Source question and definitions

The source is O. F. K. Kalenda and J. Spurny, *On quantitative Schur and Dunford–Pettis properties*, arXiv:1302.6369v2 [1]. On source page 12, Example 3.3 defines the spaces below and Question 3.5(1) asks:

> Is the space Y_2 from Example 3.3 hereditarily Dunford–Pettis?

The full source context is reproduced in Figure 1 at the end of the packet.

For a bounded real sequence $x = (x_k)$ and $p \in \mathbb{N}$, set

$$A_p(x) = \sup\left\{\left|x_p + x_{i_1} + \cdots + x_{i_p}\right| : p < i_1 < \cdots < i_p\right\}.$$

For each $n \in \mathbb{N}$, the source defines

$$X_n = (c_0, \|\cdot\|_n), \qquad \|x\|_n = \max_{1\le p\le n} A_p(x),$$

and

$$Y_2 = \left(\bigoplus_{n=1}^{\infty} X_n\right)_{c_0}.$$

It also recalls the Pełczyński–Szlenk space

$$Y_1 = \{x \in \ell_\infty : A_p(x) \to 0\}, \qquad \|x\|_{Y_1} = \sup_p A_p(x).$$

Example 3.3 states that Y_1 fails the Dunford–Pettis property: its canonical basis (e_k) and biorthogonal functionals (e_k^*) are both weakly null while $e_k^*(e_k) = 1$ [1, 2].

2 Proof intuition

The norm of the pth summand X_p sees only $A_1, \dots, A_p$. To isolate the new information at scale p, erase all coordinates before p. The resulting tail has X_p-norm comparable to $A_p(x)$ alone. Indeed, every shorter sum in that tail can be completed with far-out coordinates tending to zero to become an A_p sum. It costs at most one additional copy of $|x_p|$, itself bounded by $A_p(x)$. Thus placing the pth tail into the pth summand produces a diagonal copy of Y_1 in Y_2. Since the component norms are controlled by $A_p(x) \to 0$, the diagonal really belongs to the c_0-sum.

3 Tail estimate and solution

For $p \in \mathbb{N}$, define

$$P_p x = (\underbrace{0, \dots, 0}_{p-1 \text{ entries}}, x_p, x_{p+1}, \dots).$$

Lemma 1 (Tail comparison)**.** *For every $x \in Y_1$ and $p \in \mathbb{N}$,*

$$A_p(x) \le \|P_p x\|_p \le 2A_p(x). \tag{1}$$

Moreover, every $x \in Y_1$ belongs to c_0.

Proof. First, $|x_p| \le A_p(x)$: in the definition of $A_p(x)$ choose all p tail indices tending to infinity. Similarly, for every $k > p$, choose one tail index equal to k and the other $p-1$ indices tending to infinity. This gives

$$|x_p + x_k| \le A_p(x), \qquad |x_k| \le 2A_p(x).$$

Since $A_p(x) \to 0$, it follows that $x \in c_0$.

Clearly $A_p(P_p x) = A_p(x)$, which proves the lower bound in (1). Fix $q < p$ and a sum occurring in the definition of $A_q(P_p x)$. After deleting its zero terms, it is a sum over a set $E \subseteq \{p, p+1, \dots\}$ with $|E| \le q$.

If $p \in E$, complete $E \setminus \{p\}$ to p indices strictly larger than p by adding indices tending to infinity. The corresponding A_p sums converge to $\sum_{j \in E} x_j$, so its absolute value is at most $A_p(x)$. If $p \notin E$, the same completion gives

$$\left| x_p + \sum_{j \in E} x_j \right| \le A_p(x).$$

Together with $|x_p| \le A_p(x)$, this bounds the original sum by $2A_p(x)$. Hence $A_q(P_p x) \le 2A_p(x)$ for every $q < p$, while the $q = p$ term equals $A_p(x)$. Taking the maximum proves the upper bound. □

Theorem 1. *The answer to Question 3.5(1) is negative: Y_2 is not hereditarily Dunford–Pettis. More precisely, Y_2 contains a closed subspace two-isomorphic to Y_1.*

Proof. Define the linear map

$$J : Y_1 \longrightarrow \prod_{p=1}^{\infty} X_p, \qquad Jx = (P_p x)_{p=1}^{\infty}.$$

By Lemma 1,

$$\|P_p x\|_p \le 2A_p(x) \longrightarrow 0,$$

so $Jx \in Y_2$. The same lemma yields

$$\|x\|_{Y_1} = \sup_p A_p(x) \le \|Jx\|_{Y_2} \le 2 \sup_p A_p(x) = 2\|x\|_{Y_1}.$$

Thus J is an isomorphic embedding with closed range. That range is isomorphic to Y_1, which fails the Dunford–Pettis property by Example 3.3 of the source. Therefore Y_2 has a closed subspace without the Dunford–Pettis property and is not hereditarily Dunford–Pettis. □

4 Verification, novelty, and limitations

The accompanying script checks (1) on 28,000 finitely supported random vector/tail pairs. This is a bookkeeping check, not a proof. A bounded search on 11 August 2026 used the exact question, the source title, the phrases "Y_2 hereditarily Dunford–Pettis" and "c_0-sum of the spaces X_n", and close tail-embedding terms. The 2015 journal version still states the question, and no later explicit answer or this embedding was found. Novelty confidence is moderate because the argument is elementary and may be unindexed folklore.

This packet answers only Question 3.5(1). It does not settle Questions 3.4, 3.5(2), or 3.5(3). In particular, Theorem 1 shows that Y_2 cannot serve as a hereditarily Dunford–Pettis counterexample to Question 3.5(2).

Human-review recommendation. Recommended for expert review as a complete negative solution. The key checks are the filler-index limit in Lemma 1 and the bounded novelty status.

References

[1] O. F. K. Kalenda and J. Spurny, *On quantitative Schur and Dunford–Pettis properties*, Bull. Aust. Math. Soc. 91 (2015), 471–486, arXiv:1302.6369.

[2] A. Pełczyński and W. Szlenk, *An example of a non-shrinking basis*, Rev. Roumaine Math. Pures Appl. 10 (1965), 961–966.

Example 3.3. *There are Banach spaces Y_1 and Y_2 with the following properties:*

(i) *Both Y_1 and Y_2 are isometric to subspaces of $C[0, \omega^\omega]$.*
(ii) *Y_1 fails the Dunford-Pettis property, so Y_1^* fails the Schur property.*
(iii) *Y_2^* has the Schur property, so Y_2 has the dual quantitative Dunford-Pettis property. Y_2 fails the direct quantitative Dunford-Pettis property*

Proof. The existence of Y_1 is a result of [13]. The above spaces X_n are in a sense approximations of Y_1. For the sake of completeness let us recall the definition of Y_1.

The quantity $A_p(x)$ defined above has sense for any $x \in \mathbb{R}^{\mathbb{N}}$. Furher, it is finite if and only if x is bounded. The space $(Y_1, \|\cdot\|)$ is defined as

$$Y_1 = \{x \in \ell_\infty : A_p(x) \to 0\}, \qquad \|x\| = \sup\{A_p(x) : p \in \mathbb{N}\}.$$

It is proved in [13] that Y_1 is isometric to a subspace of $C[0, \omega^\omega]$, the canonical basis (e_k) of Y_1 is unconditional, the orthogonal functionals (e_k^*) form also an uncoditional basis of Y_1^*, e_k weakly converge to zero, e_k^* as well, while $e_k^*(e_k) = 1$. This proves the failure of the Dunford-Pettis property.

The space Y_2 can be taken to be the c_0-sum of the spaces X_n, $n \in \mathbb{N}$. Then all the properties easily follow. □

In view of the previous example the following question seems to be natural.

Question 3.4. *Let X be a Banach space isometric to a subspace of $C(K)$ with K scattered. Suppose that X has the Dunford-Pettis property. Does X^* have the Schur property? Does X have the dual quantitative Dunford-Pettis property?*

A related topic is the study of spaces having hereditary Dunford-Pettis property, i.e., spaces all whose subspaces enjoy the Dunford-Pettis property. Within $C(K)$ spaces they are exactly those such that K has finite height (as explicitely formulated in [3, Theorem 1] as a consequence of [13]). Further, spaces with the Schur property enjoy hereditary Dunford-Pettis property as well. Further, the space constructed by Hagler in [6] has also hereditary Dunford-Pettis property by [3, Proposition 2]. It seems to us that the following questions are interesting.

Question 3.5.

(1) *Is the space Y_2 from Example 3.3 hereditarily Dunford-Pettis?*

Figure 1: Source page 12: Example 3.3 and Question 3.5(1).